

AI Intelligence Graph — Architecture

1. Ingestion

Public startup/product listings, arXiv research, RSS news, and public job JSON endpoints are collected with bounded timeouts. Source URLs are retained for traceability.

2. Reliability

HTTP failures are isolated per source. LLM calls use provider fallback, exponential backoff with jitter for HTTP 429, and smaller chunks after HTTP 413.

3. Freshness

News and jobs are admitted only when their source timestamp falls within the exact 24-hour window (with a small future clock-skew allowance). Missing timestamps are rejected.

4. Entity resolution

Names are normalized and compared deterministically against a 50-record seed list using similarity scoring. Every mapping retains the source URL.

5. Storage

CSV files are Google-Sheet-ready; an XLSX workbook contains the six required tabs. A production deployment can move raw documents to object storage, primary records to PostgreSQL, vectors to a vector index, and relationships to a graph database.

Layer	Demo implementation	Production direction
Sources	HTTP/RSS/JSON	Async workers + queue
Primary records	CSV/XLSX	PostgreSQL
Vectors	Not required for demo	Vector DB
Relationships	Entity Mapping Log	Graph DB
Observability	Console diagnostics	Metrics/traces/alerts